DR. EMILY HATT

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Publications

Postdoctoral researcher in the field of Asteroseismology. Scientific expertise includes generating an automated pipeline to detect solar-like oscillations, using asteroseismology to measure core rotation and magnetism, employing hierarchical Bayesian modelling and emulating grids of stellar models using a neural network. Proficient software developer, especially skilled in Python.

SCIENTIFIC POSITIONS

Postdoctoral Researcher, ISTA — *June 2026 - present*
Prof. Lisa Bugnet

Research Fellow, University of Birmingham — *Jul 2024 - May 2026*
Prof. William Chaplin, Prof. Guy Davies

PhD Astrophysics, University of Birmingham — *Sep 2020 - Jun 2024*
Supervised by Prof. William Chaplin
Asteroseismic Inference on a Population Scale: From Detection to Internal Properties

EDUCATION

University of Birmingham, MSci Physics + Astrophysics — *Sep 2016 - Jul 2020*
Result: 1st Class Honours

The Sixth Form College Farnborough — *Sep 2014 - Jul 2016*

KEY PUBLICATIONS

See QR code at the top of this page for a full list of publications.

Hatt, E., J. M. J. Ong, M. B. Nielsen, *et al.*, "Asteroseismic signatures of core magnetism and rotation in hundreds of low-luminosity red giants,", vol. 534, no. 2, pp. 1060–1076, Oct. 2024. DOI: `10.1093/mnras/stae2053`. arXiv: `2409.01157 [astro-ph.SR]`

Hatt, E., M. B. Nielsen, W. J. Chaplin, *et al.*, "Catalogue of solar-like oscillators observed by TESS in 120-s and 20-s cadence,", vol. 669, A67, A67, Jan. 2023. DOI: `10.1051/0004-6361/202244579`. arXiv: `2210.09109 [astro-ph.SR]`

M. B. Nielsen, J. M. J. Ong, **Hatt, E. J.**, *et al.*, "Asteroseismology with PBjam 2.0: Measuring Dipole Mode Frequencies in Coupling Regimes from Main-sequence to Low-luminosity Red Giant Stars,", vol. 169, no. 6, 322, p. 322, Jun. 2025. DOI: `10.3847/1538-3881/adcb37`. arXiv: `2506.20382 [astro-ph.SR]`

J. Ong, M. Nielsen, **Hatt, E.**, *et al.*, "Reggae: A Parametric Tuner for PBJam, and a Visualization Tool for Red Giant Oscillation Spectra," *The Journal of Open Source Software*, vol. 9, no. 99, 6588, p. 6588, Jul. 2024. DOI: `10.21105/joss.06588`. arXiv: `2407.09956 [astro-ph.IM]`

M. B. Nielsen, **Hatt, E.**, W. J. Chaplin, *et al.*, "A probabilistic method for detecting solar-like oscillations using meaningful prior information. Application to TESS 2-minute photometry,", vol. 663, A51, A51, Jul. 2022. DOI: `10.1051/0004-6361/202243064`. arXiv: `2203.09404 [astro-ph.SR]`

W. H. Ball, A. H. M. J. Triaud, **Hatt, E.**, *et al.*, "Projected spin-orbit alignments from Kepler asteroseismology and Gaia astrometry,", vol. 521, no. 1, pp. L1–L4, Jun. 2023. DOI: `10.1093/mnrasl/slad012`. arXiv: `2301.10308 [astro-ph.SR]`

M. B. Nielsen, G. R. Davies, incl **Hatt, E**, *et al.*, "Simplifying asteroseismic analysis of solar-like oscillators. An application of principal component analysis for dimensionality reduction,", vol. 676, A117, A117, Aug. 2023. DOI: `10.1051/0004-6361/202346086`. arXiv: `2306.13577 [astro-ph.SR]`

M. J. Goupil, C. Catala, incl **Hatt, E**, *et al.*, "Predicted asteroseismic detection yield for solar-like oscillating stars with PLATO,", vol. 683, A78, A78, Mar. 2024. DOI: `10.1051/0004-6361/202348111`. arXiv: `2401.07984 [astro-ph.SR]`

TALKS

Cool Stars 23 (conference) *2026, invited*
Recent Observational Advances insights into Post Main-Sequence Rotation and Magnetism Review talk for splinter session [20 mins]

CFSA Series, Warwick *2026, invited*
Seeing into Stars with Asteroseismology and Advanced Statistics - The Missing Angular Momentum Transport Problem.

ISTA Astronomy Seminar Series *2025, invited*
Seeing into Stars with Asteroseismology and Advanced Statistics
Talk given to the Astronomy department at the Institute of Science and Technology, Austria [1 hour]

Asteroseismology for Exoplanet Scientists *2025, internal*
Asteroseismology: Fundamentals and Applications
Talk given to University of Birmingham research group. [1 hour]

Astronomy on Tap *2025, invited*
Dancing with the Stars: Stars, Sound and Me.
Public outreach talk about stellar physics and asteroseismology.

Tea, Talk and Telescope *2024, invited*
Stars, Sound and Me
Seminar given to University of Birmingham Astronomical Society. [1 hour]

Tutorial on the use of Canva for presentations *2024, internal*
Giving memorable presentations; from content to visuals with Canva
Session delivered to members of University of Birmingham research group presenting tips and tricks for giving memorable presentations, and how to use the design tool *Canva* to make interesting slides [1 hour]

Asteroseismic Inference on a Population Scale *2024, internal*
Overview of my PhD
Talk given to University of Birmingham research group. [1 hour]

Asteroseismology Tutorial *2023, internal*
Introduction to Asteroseismic Data Analysis

Session delivered to members of University of Birmingham research group dedicated to introducing asteroseismology and walking attendees through some basic analysis tools, including `pbjam` and `Lightkurve`. [1 hour]

TASC7/KASC14 (conference) *2023*
An Ensemble View of Magnetic Imprints on Mixed Modes at the Base of the RGB

TESS mission Update Meeting (conference) *2023, invited*
A Catalogue of Solar-Like Oscillators in Short Cadence

SASP EDI Event *2022, organised*
Space is Black: A Celebration of Black Astrophysicists
Event marking the unveiling of a set of posters I lead in designing to promote inclusivity. Gave a talk about the status of inclusivity in Astronomy.

Asterochronometry Seminar Series *2022, invited*
A probabilistic method for detecting solar-like oscillations and results in TESS data
Virtual seminar for the Asterochronometry group at the University of Bologna. [30 mins]

TASC6/KASC13 (conference) *2022*
Asteroseismology with TESS Short Cadence Data: Finding thousands of new solar-like oscillators using a probabilistic detection algorithm

Good Vibrations Seminar Series *2021, invited*
Asteroseismology with TESS; Bridging the Gap Between The Red Giant Branch and the Main Sequence
Virtual seminar hosted by Universitè PSL. Presentation viewable online. [45 mins]

National Astronomy Meeting (conference) *2021*
TESS Asteroseismology: New Detections from Main-sequence to Subgiant Stars

TEACHING

Summer Studentship: Physics and Art *Sept 2025*
Supervised a summer studentship project combining art and astrophysics.

Undergraduate 1st Year Astrolab *2025*
Guided students through astrophysical laboratory projects and contributed to developing the laboratory manual.

Undergraduate 3rd Year Group Studies *2023*
Supervised a group project concerning the asteroseismic potential of PLATO. Headed the modelling team, consisting of 9 students.

Undergraduate 1st Year Mathematics Problem Classes *2021 - 2023*
Guided students through mathematics problems. Involved both one-to-one support and demonstrating solutions to the class.

Undergraduate 2nd Year Computing Lab *2021 - 2023*
Supported students in learning how to code in Python. Introduced the Jupyter Notebook in the process.

OUTREACH

Offer Holder Visit Days *2022 - 2023*
Ran 45 minute Astrophysics workshops held for students with offers for undergraduate study at the University of Birmingham. Attended on average one per semester.

Workshops for Home Educated Students *2022 - 2023*
Aided in running educational workshops on campus for home educated students.

Lead Organiser of E&D Committee *2021 - 2023*

Headed organisation of the Birmingham Sun, Stars and Exoplanets Equality and Diversity Committee. Lead project producing posters highlighting astronomers from underrepresented groups. Organised an unveiling event, with an invited speaker. Aided in developing several school workshops aimed at increasing inclusion in astronomy, which were delivered at a number of events.